Convergent Loss of CMAH Across Mammalia, the Neonatal Immunological Window, and Implications for Selective Biological Targeting

Research Document Nuwan Ambagahawatte Thompson Rivers University, Kamloops, British Columbia October 2026

Abstract

This document surveys three intersecting areas of biology: (1) the convergent inactivation of the CMAH gene across mammalian lineages and its consequences for cell-surface sialic acid profiles; (2) the transient immunological window in human neonates defined by the simultaneous presence of fetal hemoglobin, maternal immunoglobulin G, and a milk oligosaccharide-dominated gut microbiome; and (3) the theoretical implications of these biological signatures for selective biochemical targeting. A review of atmospheric residence times for biological aerosols and self-neutralizing atmospheric chemistry is included. All claims are sourced from peer-reviewed literature.

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1. CMAH Gene Inactivation and Neu5Gc Loss in Humans

1.1 The Human-Specific Loss

Approximately 2-3 million years ago, the human lineage suffered a 92 base-pair deletion in the CMAH gene, which encodes the enzyme CMP-Neu5Ac hydroxylase. This enzyme converts N-acetylneuraminic acid (Neu5Ac) to N-glycolylneuraminic acid (Neu5Gc), a sialic acid displayed on the surface of virtually all mammalian cells. The deletion rendered the gene non-functional, making humans unable to synthesize Neu5Gc. All other great apes — chimpanzees, gorillas, orangutans — and all Old World monkeys retain functional CMAH and continue to produce Neu5Gc.

The loss is considered one of the defining biochemical distinctions between humans and their closest evolutionary relatives. Human immune systems treat Neu5Gc as a foreign antigen, producing circulating anti-Neu5Gc antibodies — a phenomenon with implications for xenotransplantation, red meat consumption, and cancer biology.

Sources: - Varki A. "Loss of N-glycolylneuraminic acid in humans: Mechanisms, consequences, and implications for hominid evolution." *American Journal of Physical Anthropology* 116(S33): 54-69, 2001. - Chou HH et al. "A mutation in human CMP-sialic acid hydroxylase occurred after the Homo-Pan divergence." *PNAS* 95(20): 11751-11756, 1998. - Okerblom J & Varki A. "Biochemical, Cellular, Physiological, and Pathological Consequences of Human Loss of N-Glycolylneuraminic Acid." *ChemBioChem* 18(13): 1155-1171, 2017.

1.2 Convergent CMAH Loss Across Mammalia

While often described in the literature as a "human-specific loss," the inactivation of CMAH has occurred independently in multiple mammalian lineages — a striking example of convergent molecular evolution. A comprehensive phylogenetic survey by Okerblom et al. (2017), building on Springer et al. (2014), identified at least 31 independent losses of CMAH during deuterostome evolution. Among mammals, the following lineages have confirmed loss of functional CMAH and absence of Neu5Gc:

Lineage	Species Count	Estimated Timing	Mechanism
Hominins (Homo)	1 species	~2-3 Mya	92 bp deletion
New World monkeys (Platyrrhini)	100+ species	~30 Mya	Inversion of exons 4-13; loss of exons 4-8 and 10-13
Mustelidae (ferrets, martens, weasels, otters)	57 species	Variable	Multiple independent events
Pinnipedia (seals, sea lions, walruses)	33 species	Variable	Gene inactivation
Procyonidae (raccoons, coatis, ringtails)	~15 species	Variable	Gene inactivation
Erinaceidae (hedgehogs)	17 species	Variable	Gene inactivation
Chiroptera (bats) — two lineages	Variable	Independent in Vespertilionidae and Rhinolophidae/ Hipposideridae	Two separate inactivation events; CMAH retained in Pteropodidae
White-tailed deer (*Odocoileus virginianus*)	1 species	Unknown	Gene inactivation; all other cervids retain CMAH

Lineage	Species Count	Estimated Timing	Mechanism
Sperm whale (*Physeter macrocephalus*)	1 species	Unknown	Gene inactivation; all other cetaceans retain CMAH

Notable observations:

- The losses are phylogenetically scattered. There is no pattern by body size, habitat, or diet. A hedgehog and a sperm whale share the same molecular quirk by entirely unrelated genetic accidents.
- Among cetaceans, only the sperm whale — possessing the largest brain ever to exist on Earth (approximately 7.8 kg, six times the mass of the human brain) — has lost CMAH. All dolphins, baleen whales, and other toothed whales retain functional CMAH.
- Among cervids (deer), only the white-tailed deer has lost CMAH. All other deer species retain it.
- The great apes are entirely unaffected. Chimpanzees, bonobos, gorillas, and orangutans all retain functional CMAH and produce Neu5Gc normally.

The functional consequences of CMAH loss vary by lineage. In ferrets, the absence of Neu5Gc results in "naturally humanized" influenza A virus receptors, making ferrets the preferred animal model for human influenza research (Ng et al., 2014). In New World monkeys, the loss occurred approximately 30 million years ago and has had no apparent detrimental effect on species fitness.

Sources: - Okerblom JJ et al. "Loss of CMAH during Human Evolution Primed the Monocyte-Macrophage Lineage toward a More Inflammatory and Phagocytic State." *Journal of Immunology* 198(6): 2366-2373, 2017. PubMed - Okerblom JJ et al. "Biochemical, Cellular, Physiological, and Pathological Consequences of Human Loss of N-Glycolylneuraminic Acid." *ChemBioChem* 18(13): 1155-1171, 2017. Wiley - Springer SA & Bhatt S. "Parallel evolution of a self-signal: humans and new world monkeys independently lost the cell surface sugar Neu5Gc." *Immunogenetics* 67: 191-202, 2014. Springer - Ng PSK et al. "Ferrets exclusively synthesize Neu5Ac and express naturally humanized influenza A virus receptors." *Nature Communications* 5: 5750, 2014. Nature - Varki A. "Colloquium paper: uniquely human evolution of sialic acid genetics and biology." *PNAS* 107(Suppl 2): 8939-8946, 2010. PMC

1.3 Neu5Gc as a Biochemical Species Marker

The absence of Neu5Gc on human cell surfaces, combined with the presence of circulating anti-Neu5Gc antibodies, constitutes the cleanest species-level biochemical marker in mammalian biology. It is:

- Binary (present or absent; no gradient)
- Universal within the species (all humans lack Neu5Gc; no known exceptions)
- Stable (the gene has been non-functional for over 2 million years)
- Detectable at the cellular level via lectin staining, mass spectrometry, or antibody assays

However, as documented above, it is not unique to humans. Any biochemical system keyed to Neu5Gc absence as a proxy for "human" would also identify New World monkeys, mustelids, pinnipeds, procyonids, hedgehogs, specific bat lineages, white-tailed deer, and sperm whales as positive matches.

This has practical implications for xenotransplantation research, where pig organs engineered to lack Neu5Gc (via CMAH knockout) are being developed for human transplantation. It also has implications for any theoretical selective biological agent designed to target or spare organisms based on their sialic acid profile.

Sources: - Salama A et al. "Anti-Neu5Gc antibodies in humans." *Xenotransplantation* 22(4): 244-251, 2015. PMC - Dhar C et al. "Thresholds of Glycan-Mediated Xenoreactivity: Anti-Neu5Gc Antibodies and Their Implications for Xenotransplantation." *Glycobiology* 32(9): 791-802, 2022. Oxford

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2. The Neonatal Immunological Window

2.1 Overview

Human neonates occupy a transient immunological state defined by the simultaneous presence of three factors not found together in any other age group:

1. **Fetal hemoglobin (HbF)** — a gamma-globin-based hemoglobin variant dominant at birth
2. **Maternal immunoglobulin G (IgG)** — passively transferred across the placenta in the third trimester
3. **A gut microbiome dominated by Bifidobacterium species sustained by human milk oligosaccharides (HMOs)**

Each factor decays on its own timeline. The period in which all three are simultaneously present at significant levels constitutes a unique biological window.

2.2 Factor 1: Fetal Hemoglobin

Fetal hemoglobin (HbF, alpha-2/gamma-2) constitutes 60-80% of total hemoglobin at birth. The "hemoglobin switch" — the transition from gamma-globin to beta-globin production — begins near term and proceeds over the first year of life:

Age	Approximate HbF (%)
Birth	60-80%
1 month	50-70%
3 months	20-30%
6 months	2-10%
12 months	1-2%
Adult	<1%

HbF persists above adult baseline levels until approximately 12 months, though the most significant decline occurs between 2-6 months.

Hereditary Persistence of Fetal Hemoglobin (HPFH): A group of benign genetic conditions in which HbF production continues into adulthood. Prevalence varies by population: approximately 0.1% in African populations, with heterocellular forms (HbF 1-4%) affecting up to 10% of some populations. Homozygous deletional HPFH can produce near-100% HbF in adults, though this is extremely rare (estimated <1 in 500,000 globally).

Pharmacological induction: Hydroxyurea, used in sickle cell disease management, raises adult HbF to a mean of approximately 17%, with top responders reaching 18-20%. This is comparable to infant levels at approximately 4 months of age.

Sources: - Sankaran VG & Orkin SH. "The switch from fetal to adult hemoglobin." *Cold Spring Harbor Perspectives in Medicine* 3(1): a011643, 2013. - Forget BG. "Molecular basis of hereditary persistence of fetal hemoglobin." *Annals of the New York Academy of Sciences* 850: 38-44, 1998. PMC - Steinberg MH et al. "Fetal hemoglobin in sickle cell anemia." *Blood* 89(3): 1078-1088, 1997. ASH

2.3 Factor 2: Maternal IgG

Immunoglobulin G is the only antibody class that crosses the human placenta. Transfer occurs primarily in the third trimester via the neonatal Fc receptor (FcRn), with the majority of transfer after 28 weeks gestation.

Decay in the neonate is biphasic: - Initial rapid phase: half-life approximately 22-23 days (first 30 days) - Subsequent slower phase: half-life approximately 37-38 days

By practical clinical assessment: - 3-4 months: maternal antibodies to many pathogens have declined to non-protective levels - 6-9 months: most maternal IgG is undetectable for most antigen specificities - 12 months: only trace amounts persist (primarily measles-specific IgG in infants of naturally immune mothers)

No biological mechanism exists for maternal IgG to persist into adulthood. The molecule has a circulating half-life of approximately 21 days and is not regenerated once the placental supply ceases. This factor is absolutely infant-specific and cannot be present in any adult under natural conditions.

Sources: - Palmeira P et al. "IgG placental transfer in healthy and pathological pregnancies." *Clinical and Developmental Immunology* 2012: 985646, 2012. PMC - Niewiesk S. "Maternal antibodies: clinical significance, mechanism of interference with immune responses, and possible vaccination strategies." *Frontiers in Immunology* 5: 446, 2014. - Leuridan E et al. "Early waning of maternal measles antibodies in era of measles elimination." *BMJ* 340: c1626, 2010.

2.4 Factor 3: HMO-Dominated Gut Microbiome

In breastfed infants, human milk oligosaccharides (HMOs) — complex sugars indigestible by the infant — serve as the primary carbon source for specific gut bacteria, particularly Bifidobacterium longum subsp. infantis, which possesses the complete enzymatic machinery to metabolize HMOs.

The resulting gut profile: - Bifidobacterium dominance: approximately 50% at 2 weeks, reaching up to 75% by 6 months in breastfed infants - Substantial individual variation exists (0.001% to 91% across infants) - Approximately 25% of US infants have no detectable Bifidobacterium, even among vaginally delivered, breastfed infants - The HMO-dominated profile ends at weaning (cessation of breastfeeding, typically between 6-24 months)

Adults do not naturally harbour B. infantis-dominated gut flora. A study in Cell Host & Microbe demonstrated that B. infantis can temporarily engraft in adult guts when co-administered with HMOs as a synbiotic, but de-engrafts upon cessation of HMO supplementation. No natural adult condition produces this microbiome profile.

Sources: - Henrick BM et al. "Bifidobacteria-mediated immune system imprinting early in life." *Cell* 184(15): 3884-3898, 2021. - Samara J et al. "Supplementation with a probiotic mixture accelerates gut microbiome maturation and reduces intestinal inflammation in extremely preterm infants." *Cell Host & Microbe* 30(5): 696-711, 2022. Cell - Bokulich NA et al. "Antibiotics, birth mode, and diet shape microbiome maturation during early life." *Science Translational Medicine* 8(343): 343ra82, 2016. - Underwood MA et al. "Bifidobacterium longum subspecies infantis: champion colonizer of the infant gut." *Pediatric Research* 77: 229-235, 2015. PMC

2.5 The Three-Factor Overlap Window

Mapping the simultaneous presence of all three factors:

Factor	Significant Presence	Declining	Trace/Absent
Fetal hemoglobin	0-4 months	4-8 months	8-12 months (residual 2-5%)
Maternal IgG	0-3 months	3-6 months	6-12 months (trace)
HMO gut flora	0 to weaning	At weaning	Weeks post-weaning

The window in which all three factors are robustly present is approximately **0-4 months**. The window in which all three are detectable, even at residual levels, extends to approximately **8-10 months** in breastfed infants — contingent on continued breastfeeding for the third factor.

This three-factor state has no equivalent in adult biology. It is unique to the neonatal period and defined by developmental processes (hemoglobin switching, passive antibody decay, weaning) that are irreversible under natural conditions.

Significance: The three-factor window represents a transient biological signature that could, in principle, serve as a binary discriminator between neonates and all other human age groups. Unlike age-dependent biomarkers that decline gradually, the conjunctive requirement (all three simultaneously) creates a sharp boundary: present or absent, with minimal grey zone.

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3. Atmospheric Dispersal of Biological Aerosols

3.1 Residence Times

The behaviour of biological particles in the troposphere has been extensively studied in the context of bioaerosol ecology and bioweapon defence:

Particle Type	Atmospheric Residence Time	Transport Distance
Bacteria (1-5 microns)	Hours to ~1 week	Hundreds to thousands of km
Fungal spores	Days to ~2 weeks	Transcontinental
Viruses (<0.3 microns)	Potentially >1 week	Global (theoretical)
Fine aerosol (<2.5 microns)	Days to weeks	Hemispheric

Removal is primarily by precipitation (wet deposition) and gravitational settling (dry deposition). Smaller particles remain airborne longer and travel further.

3.2 Global Dispersal Timescales

- **Zonal mixing** (east-west within one hemisphere): approximately 2-4 weeks
- **Meridional mixing** (north-south within one hemisphere): approximately 1-3 months
- **Interhemispheric exchange**: approximately 4 months to 1 year
- **Stratospheric injection** (if particles reach above the tropopause): residence times of 1-3 years, but injection itself requires volcanic eruption-scale energy or deliberate high-altitude release

A biological agent released from a single point source cannot achieve global coverage in less than several weeks. Uniform global distribution within 48 hours would require simultaneous release from hundreds of geographically distributed points across both hemispheres, or an entirely novel atmospheric delivery mechanism.

3.3 Self-Neutralizing Atmospheric Chemistry

The troposphere possesses a natural self-cleaning capacity mediated primarily by the hydroxyl radical (OH), which oxidizes and neutralizes trace gases and organic aerosols. This system processes approximately 3.7 billion tonnes of reactive trace gases annually.

Engineered compounds can be designed with built-in degradation pathways: - **UV photolysis**: compounds that decompose upon exposure to sunlight (many chemical warfare agents degrade via this pathway) - **Hydrolysis**: compounds that neutralize upon contact with atmospheric water vapour - **Radical-initiated decomposition**: compounds designed to react with naturally present OH radicals after a predetermined atmospheric residence time

A "second release" of a neutralizing compound that specifically targets and decomposes a first compound is theoretically grounded in atmospheric remediation chemistry, though no large-scale

deployment has been attempted. The primary challenge is matching the dispersal and residence characteristics of the target compound.

Sources: - Despres VR et al. "Primary biological aerosol particles in the atmosphere: a review." *Tellus B* 64: 15598, 2012. Tandfonline - Frohlich-Nowoisky J et al. "Bioaerosols in the Earth system: Climate, health, and ecosystem interactions." *Atmospheric Research* 182: 346-376, 2016. ScienceDirect - Reche I et al. "Deposition rates of viruses and bacteria above the atmospheric boundary layer." *ISME Journal* 12: 1154-1162, 2018. Oxford - Lelieveld J et al. "The Self-Cleansing Capacity of the Troposphere." *Atmospheric Chemistry and Physics* 4: 2337-2344, 2004.

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4. Implications for Selective Biological Targeting

4.1 Sialic Acid Profile as a Targeting Mechanism

The Neu5Gc/Neu5Ac distinction offers a theoretically precise targeting mechanism. A compound designed to interact specifically with the Neu5Ac-only cell-surface profile (characteristic of humans and the mammalian lineages documented in Section 1.2) would:

- Affect all humans
- Affect all New World monkeys (100+ species)
- Affect all mustelids (57 species)
- Affect all pinnipeds (33 species)
- Affect all procyonids (~15 species)
- Affect all hedgehogs (17 species)
- Affect specific bat lineages (two independent losses)
- Affect white-tailed deer (1 species)
- Affect sperm whales (1 species)
- **Not affect** great apes (chimpanzees, gorillas, orangutans)
- **Not affect** Old World monkeys
- **Not affect** dolphins or baleen whales
- **Not affect** most other mammalian orders

This represents approximately 11% of mammalian species. The phylogenetic scatter — unrelated lineages sharing the same molecular profile through convergent evolution — would make the pattern of affected species appear random from an ecological perspective.

4.2 The Neonatal Exception

A compound additionally keyed to the three-factor neonatal signature (Section 2) could theoretically spare neonates within the affected species. The conjunctive requirement (all three factors simultaneously present) creates a narrow but biologically sharp exemption window.

The practical implication: within the human population, individuals under approximately 8-10 months of age would present a biochemical profile distinct from all older humans. This distinction is:

- Not based on genetics (every human passes through and out of the window)
- Not modifiable (no intervention can restore the three-factor state in an adult)
- Not gradable (the conjunctive requirement makes it binary in practice)

4.3 Adult Exceptions

A small number of adults could theoretically present partial mimicry of the neonatal signature through: - HPFH (hereditary persistence of fetal hemoglobin) — providing Factor 1 - Chronic IVIG therapy — providing a non-maternal but exogenous IgG profile (a loose analogue to Factor 2) - Specific probiotic regimens with HMO supplementation — providing transient Factor 3

The probability of all three occurring simultaneously in a single adult under natural conditions is vanishingly small. Even generous estimates place it at less than 1 in 10 billion. The practical number of such individuals in a global population of 9 billion would be measured in single digits to low hundreds.

References

All sources are cited inline throughout the document. Key review articles for further reading:

1. Varki A & Gagneux P. "Multifarious roles of sialic acids in immunity." *Annals of the New York Academy of Sciences* 1253: 16-36, 2012.
2. Okerblom J & Varki A. "Biochemical, Cellular, Physiological, and Pathological Consequences of Human Loss of N-Glycolylneuraminic Acid." *ChemBioChem* 18(13): 1155-1171, 2017.
3. Palmeira P et al. "IgG placental transfer in healthy and pathological pregnancies." *Clinical and Developmental Immunology* 2012: 985646, 2012.
4. Henrick BM et al. "Bifidobacteria-mediated immune system imprinting early in life." *Cell* 184(15): 3884-3898, 2021.

5. Despres VR et al. "Primary biological aerosol particles in the atmosphere: a review." *Tellus B* 64: 15598, 2012.